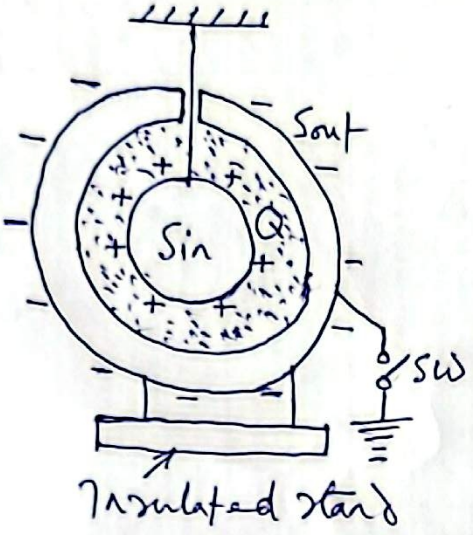
ELECTRIC FLUX DENSITY, GAUSS'S LAW AND DIVERGENCE THEOREM

Faraday's experiment



The English physicist Michael Faraday performed a very interesting experiment around 1837. First he placed a charge +Q on a small sphere Sin as shown. Then he surrounded the charged sphere by another hollow sphere Sout of larger size placed on an insulated stand. Then he filled the space btw the spheres by a dielectric and momentarily grounded the outer sphere by closing the switch (SW) so that any static charge, if present on it, is neutralized; thus, zero initial charge on it.

Finally, he removed the inner charged sphere Sin and measured the charge on the outer sphere. He found that the total charge on the outer sphere was equal in magnitude to the original charge on the inner sphere.

Faraday repeated the experiment many